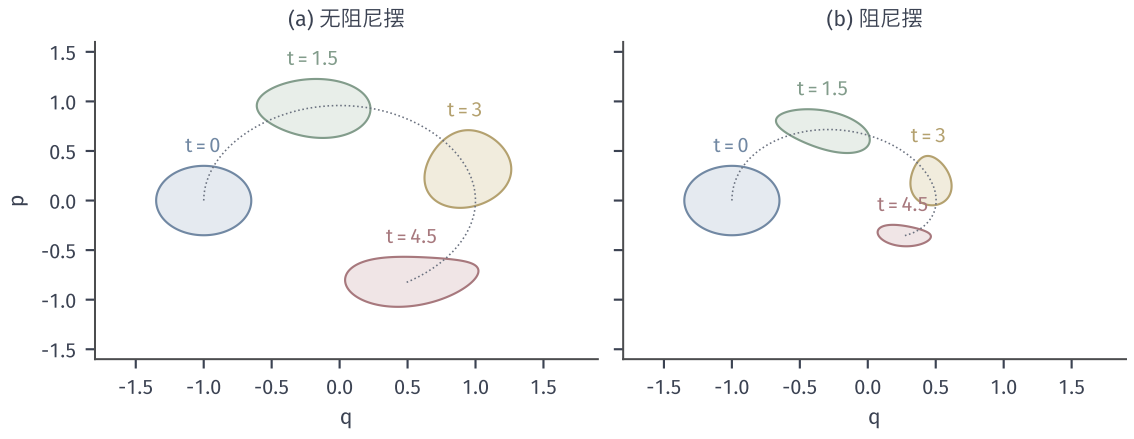
(a) 无阻尼摆
t = 1.5
t = 3
t = 0
t = 4.5
1.5
1.0
0.5
0.0
-0.5
-1.0
-1.5
p
-1.5
-1.0
-0.5
0.0
0.5
1.0
1.5
q
(b) 阻尼摆
t = 1.5
t = 3
t = 0
t = 4.5
-1.5
-1.0
-0.5
0.0
0.5
1.0
1.5
q